

ZITONG (ERIC) ZHOU

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EDUCATION

University of California, Los Angeles (UCLA)

Sep 2023 – June 2028 (Expected)

(Attending) Ph.D in Computer Science, *advised by Prof. Konstantinos Kallas*

University of Illinois at Urbana-Champaign (UIUC)

Aug 2021 – May 2023

M.Eng. in Electrical and Computer Engineering

New York University

Sep 2017 – May 2021

B.A. in Computer Science & Mathematics

RESEARCH INTERESTS

I work in the research area of software engineering. My research broadly spans software engineering, programming languages, and compilers. My goal is to build practical testing and analysis tools that help developers understand and verify critical software behaviors. In particular, I have worked on: (1) contextualized, LLM-based regression test augmentation targeting code changes; (2) synthesizing tests for optimizing compilers from LLM program snippets.

PUBLICATIONS

Change And Cover: Last-Mile, Pull Request-Based Regression Test Augmentation. Zitong Zhou*, Matteo Paltenghi*, Michael Pradel, Miryung Kim. *Equal Contribution. *International Conference on Software Engineering (ICSE)*. 2026.

Testing Optimizing Compilers with Grammar Composition Styles. Zitong Zhou, Ben Limpanukorn, Jiyuan Wang, Hong Jin Kang, Miryung Kim. [Preprint]

Fuzzing MLIR Compilers with Custom Mutation Synthesis. Ben Limpanukorn, Jiyuan Wang, Hong Jin Kang, Zitong Zhou, Miryung Kim. *International Conference on Software Engineering (ICSE)*. 2025.

AquaSense: Automated Sensitivity Analysis of Probabilistic Programs via Quantized Inference. Zitong Zhou, Zixin Huang, Sasa Misailovic. *Automated Technology for Verification and Analysis (ATVA)*. 2023.

RESEARCH EXPERIENCE

Taint Flow Risk Quantification

- Formulate the risk of taint flows flagged by static analyzers (e.g., Clang’s Static Analyzer, CodeQL) quantitatively using reachability analysis and flow sensitivity.
- Design and implement a LLVM-based instrumentation tool that computes the sensitivity of taint flows at runtime (e.g., during fuzzing).

Constrained Grammar Fuzzing of Compiler Optimizations

- Design grammar fuzzer (TargetFuzz) that constrains mutations to reconstruct composition styles—structural relations like adjacency, repetition, nesting between program constructs—that trigger compiler optimizations.
- Provide an API to define program constructs using grammar as input, and a default registry of composition styles to fuzz a new optimizing compiler out of the box.
- Evaluated on 63 LLVM & MLIR optimizations; outperform naïve grammar fuzzer by 2.2×; achieve highest coverage in 3/4 LLVM optimization categories and highest overall on MLIR; discover 14 new bugs (9 optimization bugs).

Automated Test Generation for Pull Requests (PRs) using LLM

- Design a LLM-based pipeline (ChaCo) that analyzes pull request (PR) context, retrieves test context, and generate tests to increase patch coverage.
- Retrieve test context—relevant tests, fixtures, parametrizations, data generators, helper utilities...—using a mixture of dynamic regression test selection and LLM-prompting; allow LLMs to generate well-integrated tests.
- Evaluated on 145 real-world PRs, ChaCo achieved 100% patch coverage on 30% PRs; Dynamic test context contributes to 100% coverage improvement. 12 tests were submitted, 9 merged, exposing 2 previously unknown bugs.

Approximate Sensitivity Analysis of Probabilistic Programs

- Implement a transpiler that takes a PP in Stan, PSI, or StormIR and outputs an sensitivity analyzer in *PyTorch*
- Achieve exact analysis on discrete models, and $18\times$ speedup on continuous models with $\leq 5\%$ accuracy loss

WORK EXPERIENCE

Dept. of Computer Science, UCLA

Teaching Assistant of CS32, CS130

Los Angeles, CA

Dept. of Computer Science, UCLA

Graduate Student Researcher

Los Angeles, CA

Dept. of Computer Science, UIUC

Researcher

May 2021 – Aug 2021

Urbana, IL

NYU CS 201 & 202

Grader

Spring '20 – Spring '21

New York, NY

DJI

Intern – Robotics Algorithm

Summer '18

Shenzhen, China

Implements a robotics control algorithm using behavior tree in *ROS*